

AI-Driven Instant Loan App

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Abstract

This report presents the design, implementation, and evaluation of an AI-powered microloan and investment application aimed at small business owners, gig workers, and individuals with irregular income. The app leverages AI for automated credit scoring, fraud detection, and investment advisory services. The goal is to address the challenges posed by traditional banking institutions, such as lengthy approval processes and a lack of financial guidance. This document follows structured guidelines to present the technical, financial, and market analysis of the proposed system.

1.0 Introduction

1.1 Problem Statement

Many individuals and small business owners struggle to access microloans due to the lack of a proper credit history. Traditional banks have lengthy approval processes, making it difficult for people needing funds. Additionally, individuals often lack financial guidance for investments, leading to poor money management. This project introduces an AI-powered loan app that enables quick, small-scale loans with automated credit scoring, fraud detection, and investment advisory using machine learning.

1.2 Initial Needs Statement

- Access Limitations: Many people cannot secure loans due to non-existent or limited credit history.
- Delayed Approval Processes: Traditional banks take too long to approve loans.
- Lack of Financial Guidance: People struggle with proper financial management due to the lack of investment advice.

2.0 Market/Customer/Business Need Assessment

2.1 Target Audience

- Small business owners
- Gig workers
- Students
- Individuals with irregular or unpredictable incomes

2.2 Market Demand

- Growing need for microloans: Economic uncertainty and rising living costs make quick, small-value loans necessary.
- High demand for AI-driven lending: Automated credit scoring ensures better risk assessment and inclusivity.
- Investment guidance necessity: Many individuals require assistance in managing investments effectively.

2.3 Competitive Advantage

- AI-powered credit risk assessment
- Instant loan approvals
- Personalized investment advisory

3.0 Target Specifications and Key Features

3.1 Registration Requirements

- Users must register with Aadhar, PAN, and a linked current account.

3.2 Loan Specifications

- Loan amounts range from INR 100 to INR 40,000.
- Withdrawals are processed in round figures.
- AI-based credit scoring determines loan eligibility.
- Dynamic interest rates adjust based on repayment behavior.

3.3 Investment Advisory

- Gullak feature: AI-driven investment suggestions for stock market trading.
- Automated financial advisory: Tailored recommendations based on user risk profiles.

4.0 External Search (Online Information Sources & References)

- RBI guidelines on digital lending and credit scoring.
- Financial reports on small-business loan requirements.
- Research papers on AI-based risk assessment and investment prediction models.

5.0 Benchmarking

5.1 Feature Comparison

Feature	Traditional Banks	Existing Loan Apps	AI-Driven Loan App
Loan Approval Time	2-7 days	24-48 hours	Instant AI-based decisions
Interest Rates	Fixed rates	Fixed rates	Dynamic ML-based rates
Credit Assessment	CIBIL scores	CIBIL + history	AI/ML-driven risk profiling
Investment Advisory	None	Basic savings	AI-powered insights

6.0 Business Model & Monetization

6.1 Revenue Streams

- Loan interest (20% base rate, increasing for late payments).
- Penalty charges for delayed repayments.
- Investment commission (10% from Gullak investment returns).

6.2 Premium Features

- Subscription-based financial advisory for advanced users.

7.0 Regulatory Considerations & External Benchmarks

7.1 Compliance Measures

- Adherence to RBI guidelines for digital lending.
- Full KYC compliance to protect against fraud.
- Stock market investment rules followed for financial safety.

7.2 Security & Data Protection

- GDPR & IT Act compliance.
- End-to-end encryption for user data protection.

8.0 Concept Development & Final Design

8.1 AI Model Refinement

- Continuously updating AI models based on real-world loan repayment data.
- Enhancing fraud detection and risk assessment using machine learning.

8.2 Future Expansions

- Integration with more financial instruments.

- Expanding into new markets for better accessibility.

9.0 Conclusion

The AI-driven microloan and investment app provides a modern financial solution for underserved populations. By leveraging AI for automated credit scoring, fraud detection, and investment guidance, this project ensures greater accessibility, security, and user empowerment. Future developments will focus on expanding services, refining AI models, and maintaining compliance with evolving financial regulations.